

Thirdhand cigarette smoke in an experimental chamber: evidence of surface deposition of nicotine, nitrosamines and polycyclic aromatic hydrocarbons...

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BACKGROUND: A growing body of evidence shows that secondhand cigarette smoke undergoes numerous chemical changes after it is released into the air: it can adsorb to indoor surfaces, desorb back into the air and undergo chemical changes as it ages. OBJECTIVES: To test the effects of aging on the concentration of polycyclic aromatic hydrocarbons (PAHs), nicotine and tobacco-specific nitrosamines in cigarette smoke. METHODS: We generated sidestream and mainstream cigarette smoke with a smoking machine, diluted it with conditioned filtered air, and passed it through a 6 m³ flow reactor with air exchange rates that matched normal residential air exchange rates. We tested the effects of 60 min aging on the concentration of 16 PAHs, nicotine, cotinine and tobacco-specific nitrosamines. We also measured sorption and deposition of nicotine, cotinine and tobacco-specific nitrosamines on materials placed within the flow reactor. RESULTS: We observed mass losses of 62% for PAHs, 72%, for nicotine, 79% for N-nitrosonornicotine and 80% for 4-(methylnitrosamino)-1-(3-pyridyl)-1-butanone (NNK). Extraction of cotton cloth exposed to smoke yielded nicotine and NNK. The ratio of NNK:nicotine on the exposed cloth was 10-fold higher than that in aerosol samples. CONCLUSIONS: Our data suggest that the majority of the PAHs, nicotine, cotinine and tobacco-specific nitrosamines that are released during smoking in homes and public places deposit on room surfaces. These data give an estimate of the potential for accumulation of carcinogens in thirdhand cigarette smoke. Exposure to PAHs and tobacco-specific nitrosamines, through dermal absorption and inhalation of contaminated dust, may contribute to smoking-attributable morbidity and mortality.